

CLOUD AND DEVOPS LAB

III B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT16	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100
COURSE OBJECTIVES: The course should enable the students to: 1. Apply the concepts of AWS and its cloud services. 2. Deploy and use virtual instances 3. Implement continuous integration using AWS CodeBuild. 4. Implement end-to-end continuous integration and continuous deployment (CI/CD) using AWS CodePipeline								
COURSE OUTCOMES: At the end of the course, student will be able to: 1. Deploy secured virtual instances in Amazon AWS. 2. Deploy various cloud services like S3, Databases 3. Proficiency in basic Git and GitHub commands. 4. Deploy automated applications								
LIST OF EXPERIMENTS								
WEEK-1								
Create Amazon AWS EC2 Linux instance with conceptual understanding of SSH client software protocol and keys.								
WEEK-2								
Create Amazon AWS EC2 Windows server instance with conceptual understanding of RDP (Remote Desktop Protocol).								
WEEK-3								
Create cloud storage Bucket using Amazon Simple Storage Service (S3). Perform the following operations: A. Create a folder within a S3 Bucket B. Upload content to S3 C. Change permissions to allow public access of contents. D. Set Meta Data on an S3 Bucket E. Delete an S3 Bucket and its content								
WEEK-4								
Launch and connect to an Amazon Relational Data Base (RDS) Service, Use any one from MySQL, Oracle, SQL Server and postgre SQLDataBase engines.								
WEEK-5								
Study and practice Git and GitHub commands								

WEEK-6
Install and configure Jenkins to build and deploy Java or Web Applications
WEEK-7
Install and configure Docker tool and perform Docker commands for content management.
WEEK-8
Create a Docker image and push Docker image into AWS ECR
WEEK-9
Create Repositories, Cloning, and Pushing Code Changes using AWS CodeCommit.
WEEK-10
Implement continuous integration using AWS Codebuild
WEEK-11
Automate application deployment into EC2 using AWS Code Deploy
WEEK-12
Implement CI/CD using AWS Code pipeline that automates the build, test, and deployment phases
TEXT BOOKS:
1. Thomas Eri, "Cloud Computing: Concepts, Technology & Architecture", Edition: 1st or later 2. Douglas E. Comer, "The Cloud Computing Book", Edition: 1st, 2023 3. Stephen Baron, "AWS: The Complete Beginners Guide", Kindle Edition 2020
REFERENCE BOOKS:
1. Hands-On Azure Devops: Cidc Implementation For Mobile, Hybrid, And Web Applications using Azure Devops and Microsoft Azure: CICD Implementation for ... Devops and Microsoft Azure (English Edition) Paperback – 1 January 2020. byMiteshSoni 2. Jeff Geerling, "Ansible for Devops: Server and configuration management for humans", First Edition, 2015. 3. David Johnson, "Ansible for Devops: Everything You Need to Know to Use Ansible for Devops", Second Edition, 2016. 4. MariotTsitoara, "Ansible 6. Beginning Git and GitHub: A Copenhensive Guide to Version Control, Project Management, and Teamwork for the New Developer", Second Edition, 2019. 5. https://www.jenkins.io/user-handbook.pdf 6. https://maven.apache.org/guides/getting-started/